OFFICEEASE – EMPLOYEE WORKPLACE ISSUE AND REQUEST MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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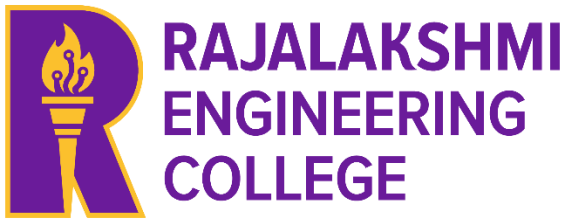
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CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

In

BACHELOR OF ENGINEERING
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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "**OFFICEEASE – EMPLOYEE WORKPLACE ISSUE AND REQUEST MANAGEMENT SYSTEM**" is the bonafide work of **Devadharshini G (240701108),** who carried out the individual project work on the Zoho Creator Low-Code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator during the academic year 2026–2027, Semester 5, under the supervision.

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I, Devadharshini G (240701108), hereby declare that the project report titled "OfficeEase – Employee Workplace Issue and Request Management System" submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its forms, reports, dashboard page, Deluge scripts and workflows — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai

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DEVADHARSHINI G

ABSTRACT

Employees in a modern office depend on many support services: IT help, building maintenance, office supplies, workstation equipment, meeting rooms, transport, training, leave, visitor handling and safety. When these requests travel through phone calls, e-mails, messaging groups and verbal instructions, details are incomplete, urgent problems are not distinguished from routine ones, the responsible team is not clearly informed, and management has no consolidated view of what is being requested or how often.

OfficeEase is a centralised low-code application built on Zoho Creator for managing employee workplace issues and requests in an office. Employees are registered once in an Employee Registration form and then submit structured requests through fifteen purpose-built forms: IT Support, Maintenance, General Workplace Issue, Office Supplies, Workstation Setup, Meeting Room and Facility booking, Transportation, Training, Attendance and Leave, Visitor Management, Workplace Safety Incident, Complaint, Feedback, and Inventory and Supplies Stock. Each form uses mandatory fields, pick lists, radio buttons, date and time pickers, number fields and file attachments so that requests are captured completely and consistently.

Deluge scripting and workflows automate the routine decisions. Ten workflows validate entries (for example, a leave end date cannot precede the start date and a supplies quantity must be greater than zero), assign a priority to general workplace issues automatically from their category, classify feedback as positive, moderate or negative from the satisfaction rating, and send e-mail alerts to the responsible team for IT support, maintenance, meeting room, safety incident and low-stock events. Fifteen list reports give each team a view of its own records, and a Dashboard page presents charts and live counters. Access is controlled through Zoho Creator profiles (Read, Write, Administrator and the Customer portal profile) and a CEO role.

The application demonstrates how a low-code platform can replace scattered, informal workplace requests with a transparent and trackable process. Features such as record-level status tracking, lookup relationships, scheduled escalation and approvals are identified in this report as future enhancements and are not represented as implemented.

Keywords: Workplace Request Management; Employee Services; Zoho Creator; Low-Code Development; Deluge Scripting; Workflow Automation; Helpdesk.

PROJECT AT A GLANCE

Attribute	Details
Project Title	OfficeEase – Employee Workplace Issue and Request Management System
Application Domain	Corporate workplace services and employee request management (Chennai office)
Student Name & Roll No.	Devadharshini G – 240701108
Development Platform	Zoho Creator low-code platform
Application Scope	Employee registration; IT support, maintenance and general issue reporting; office supplies, workstation, transport, training and leave requests; meeting room booking; visitor management; safety incidents; complaints; feedback; inventory stock; reports and dashboard
Forms	15 forms (135 data fields in total), each with mandatory-field checks, pick lists and help text
Automation	10 form workflows: 2 validation scripts, 2 on-success scripts (auto priority, feedback classification), 5 e-mail alerts and 1 placeholder workflow
Process Control	Request lifecycle: submit → validate → store → auto-priority / alert → review through reports
Reports	15 list reports (all default reports) with quick view and detail view layouts
Pages	1 dashboard page with 2 charts (column and pie) and 3 live record counters
Access Control	Profiles: Read, Write, Administrator and Customer (portal); role: CEO
External Integration Points	E-mail notifications through Zoho Creator mail (sendmail). No third-party integrations are configured
Deployment Status	Application built and exported (Version 1.0, generated on 01-Nov-2026); published application URL and production deployment evidence not supplied.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update and Delete
HR	Human Resources
IT	Information Technology
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
SLA	Service Level Agreement
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

Every workplace runs on a steady flow of small requests: a laptop that will not connect to the network, an air-conditioner that has stopped working, a box of printer toner, a meeting room for a client call, transport for a late shift, or a report of a slippery floor. Each of these needs the right team to learn about it quickly, understand exactly what is needed, judge how urgent it is and respond within a reasonable time. In many offices these requests are still made through phone calls, hallway conversations, personal e-mails and messaging groups, which makes them easy to lose and hard to track.

This project presents OfficeEase, an Employee Workplace Issue and Request Management System for a company office in Chennai, developed on the Zoho Creator low-code platform. OfficeEase brings the common employee requests, reports and feedback into a single application with structured forms, automatic validation, priority assignment, e-mail alerts to the responsible teams, reports for each service and a dashboard for management.

1.2 Problem Statement

Informal handling of workplace requests makes it difficult to know what has been requested, which requests are urgent, and which team is responsible. Missing details cause repeated follow-up, urgent safety or IT problems are not distinguished from routine requests, and management cannot see patterns such as which category of problem occurs most often.

- Requests arrive in different formats and channels, so important details such as location, urgency and employee identity are often missing.
- Urgent problems (for example, a safety hazard or a network outage) are not separated automatically from routine requests.
- Responsible teams are not notified immediately when a new request is raised.
- Leave, supplies and equipment requests contain errors (wrong dates, zero quantities) that are only found later.
- Stock levels of office supplies are not monitored, so shortages are noticed late.
- There is no consolidated view or report of requests, complaints and feedback for management.

1.3 Objectives

- Design a registration master and structured forms for each workplace service so that requests are captured completely.

- Validate entries at the time of submission to prevent invalid data.
- Assign a priority to general workplace issues automatically and classify employee feedback.
- Notify the responsible teams by e-mail when IT, maintenance, meeting room, safety or low-stock events occur.
- Provide a report for every form and a dashboard with charts and counters for quick visibility.
- Control access through Zoho Creator profiles and roles.

Objective ID	Objective	Evidence / Output
OBJ-01	Create a registration master and structured request forms for each service.	Fifteen forms in the application
OBJ-02	Capture complete, consistent data using mandatory fields, pick lists, dates and attachments.	135 data fields with help text and mandatory flags
OBJ-03	Validate entries at submission.	Leave date and supplies quantity validation workflows
OBJ-04	Prioritise issues, classify feedback and alert the responsible teams.	Automatic Issue Priority, Feedback Classification and five e-mail alert workflows
OBJ-05	Provide reports and a dashboard for live visibility.	Fifteen list reports and one Dashboard page
OBJ-06	Restrict access by profile and role.	Read, Write, Administrator, Customer profiles and CEO role

Table 1.1: Project Objectives and Measures

1.4 Scope of the Project

The project scope includes employee registration; reporting of IT, maintenance and general workplace issues; requests for office supplies, workstation setup, transportation, training and leave; meeting room and facility booking; visitor management; workplace safety incident reporting; employee complaints and feedback; inventory and supplies stock recording; reports for every form; and a dashboard page.

The following items are outside the current implementation and are discussed under future enhancements: lookup-based linking of requests to the Employee Registration record, request status and assignment fields with closure tracking, approval workflows for leave and transport, scheduled escalation of overdue requests, conditional stock alerts, record-level access restrictions, and third-party integrations such as SMS.

1.5 Justification for a Low-Code Approach

Zoho Creator provides visual form building, reports, dashboard pages, workflow configuration, e-mail actions and Deluge scripting within one platform. OfficeEase is process-oriented: most requirements involve data capture, validation, classification and notification rather than a completely custom user interface. Low-code development shortens repetitive implementation work, allows the application to be changed quickly as workplace needs evolve, and lets the developer concentrate on requirements, data design and automation logic. Deluge scripts are used where configuration alone is not enough, so the project covers the no-code to low-code range of the course.

1.6 Organisation of the Report

Chapter 2 studies the existing process and the proposed solution. Chapter 3 presents stakeholders and requirements. Chapter 4 describes architecture, data design and process design. Chapter 5 documents the forms, reports, dashboard, scripts and workflows. Chapter 6 presents testing and observations. Chapter 7 discusses deployment, security and governance. Chapter 8 reviews results and limitations, and Chapter 9 concludes with future enhancements. Appendices contain a feature checklist, a catalogue of reports, a Deluge logic summary, a user manual, the evaluation rubric and the viva-voce record sheet.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing scenario, an employee who needs IT help, a repair or a service contacts the concerned person by phone, e-mail or in person. The request is noted informally, and may or may not be forwarded to the right team. Leave applications, supplies requests and room bookings are handled on paper or in separate spreadsheets. Visitors are recorded in a register at reception, and safety incidents are reported verbally. Priorities are decided by whoever happens to receive the message, and management reports have to be prepared manually.

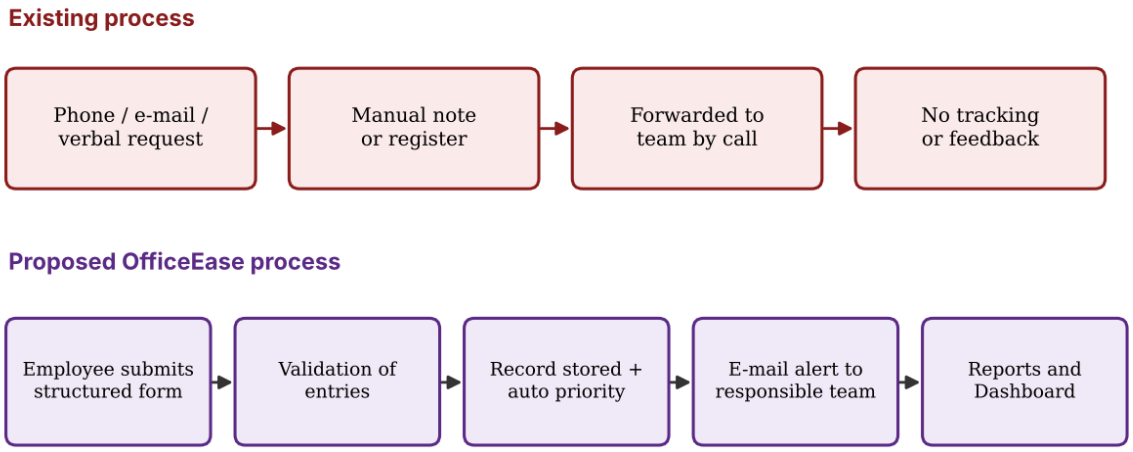


Figure 2.1: *OfficeEase request process compared with the existing process*

2.2 Limitations of the Existing System

- Requests are scattered across calls, e-mails and messages, so details are incomplete and records are lost.
- Urgent requests are not distinguished from routine ones.
- Responsible teams are not notified automatically.
- Invalid entries such as reversed leave dates or zero quantities are discovered late.
- Inventory shortages are noticed only when an item runs out.
- Management reports require manual consolidation and may be out of date.

2.3 Low-Code versus Traditional Development

Criterion	Traditional Pro-Code Development	Low-Code on Zoho Creator
Development effort	Requires a custom UI, database, APIs and deployment setup.	Uses visual form, report and page builders with configured workflows.

Criterion	Traditional Pro-Code Development	Low-Code on Zoho Creator
Data capture	Forms and validation are coded individually.	Forms, field types, mandatory flags and help text are configured visually.
Automation	Requires application logic and e-mail services to be implemented.	Workflow rules, Deluge scripts and the sendmail task support event-driven automation.
Reporting	Requires report queries and chart libraries.	List reports and chart components are generated from form data.
Access for users	Requires custom authentication and role management.	Profiles, roles and a customer portal are configurable.
Hosting	Infrastructure and release process are managed separately.	Platform hosting and application publishing are provided by the service.
Customisation	Offers broad control over architecture and UI.	Customisation is bounded by platform capabilities and plan limits.
Maintenance	Requires code-level maintenance and deployment discipline.	Routine changes are simpler, but scripts and permissions still need testing.

Table 2.1: Low-Code and Traditional Development Comparison

2.4 Literature Survey

The following sources provide the platform and process background for this project. The project brief does not include a formal research-paper survey; the table therefore focuses on authoritative platform documentation, established service-management products and a security reference rather than claiming results from experiments that were not carried out.

S.No.	Source	Focus	Relevance
1	Zoho Creator documentation [1]	Forms, reports, pages and application building	Primary reference for platform capabilities.
2	Zoho Deluge documentation [2]	Scripting language and task syntax	Supports validation and priority logic.
3	Zoho Creator workflow documentation [3]	Form-event workflow rules	Relevant to on-validate and on-success workflows.
4	Zoho Creator reports and pages documentation [4]	List reports, charts and dashboards	Relevant to reports and the Dashboard page.
5	Zoho Creator customer portal documentation [5]	Portal users and profiles	Relevant to profile-based access.
6	ServiceNow IT Service Management [6]	Enterprise request and incident management	Reference model for categorised requests and priorities.
7	Freshservice [7]	Employee service desk	Reference model for employee-facing service forms.
8	OWASP Application Security Verification Standard [8]	Application security controls	General checklist for access and data protection.

Table 2.2: Literature Survey Sources

2.5 Emerging Trends in Low-Code Development

Low-code platforms increasingly combine visual modelling, reusable components, event-driven automation, integrations and analytics. For OfficeEase, these patterns support a single operational data model, repeatable validation and automatic notifications. Automated checks should still be designed carefully: workflows need clear failure handling, role-appropriate permissions and testing with realistic data before they are trusted in daily operation.

2.6 Proposed System

The proposed system centralises workplace requests in structured Zoho Creator forms. Submitting a form triggers validation; general workplace issues receive an automatic priority from their category; feedback is classified from the satisfaction rating; and e-mail alerts are sent to the IT, maintenance, facility, safety and inventory teams. Each form has a report that lists its records, and a dashboard shows the volume of requests and the distribution of maintenance and general issues by category.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Role	Responsibilities
Employee	Register, raise IT, maintenance and general issues, request supplies, workstation equipment, transport, training and leave, book meeting rooms, report safety incidents, submit complaints and feedback.
IT Support Team	Receives IT support alerts, reviews IT Support and related issue records and resolves technical problems.
Maintenance / Facilities Team	Receives maintenance and meeting room booking alerts; reviews maintenance and workstation requests.
HR / Administration	Maintains employee registrations; reviews leave, training, complaint and transportation records.
Safety Team	Receives urgent safety incident alerts and reviews incident records.
Inventory Administrator	Maintains the inventory and supplies stock records and receives low stock alerts.
Reception / Security	Records and updates visitor entries.
Administrator / CEO	Manages application access, reviews dashboard and reports across all forms.

Table 3.1: Stakeholders and User Roles

3.2 Functional Requirements

ID	Requirement	Role(s)	Priority
FR-01	Register employees with a unique Employee ID, e-mail address and phone number, with designation, department, location, joining date, manager and status.	HR / Admin	High
FR-02	Submit IT support requests with issue type, description, priority, attachment and date reported.	Employee	High
FR-03	Submit maintenance requests with category, description, affected equipment, location, priority and attachment.	Employee	High
FR-04	Submit general workplace issues; assign the priority automatically from the issue category.	Employee / System	High
FR-05	Submit office supplies requests and reject a quantity that is zero or negative.	Employee / System	High
FR-06	Submit leave requests and reject an end date earlier than the start date.	Employee / System	High
FR-07	Submit workstation setup, transportation and training requests with the required details.	Employee	Medium
FR-08	Book meeting rooms and facilities with date, times, attendees and equipment; alert the facility team.	Employee / System	Medium
FR-09	Record workplace safety incidents with severity and injury details; alert the safety team.	Employee / System	High
FR-10	Capture employee complaints, with a confidentiality option, and employee feedback with a satisfaction rating.	Employee	Medium

ID	Requirement	Role(s)	Priority
FR-11	Classify feedback as positive, moderate or negative from the rating.	System	Medium
FR-12	Record inventory stock levels and alert the administrator by e-mail when stock is added or edited.	Inventory Admin / System	Medium
FR-13	Record visitors with purpose, person to meet, ID type and status.	Reception	Medium
FR-14	Provide a report for every form and a dashboard with charts and counters.	All roles	Medium
FR-15	Restrict access through profiles and roles.	Admin	High

Table 3.2: Functional Requirements

3.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Data integrity	Mandatory fields, unique fields in the registration form and validation scripts should keep data consistent.
NFR-02	Security and privacy	Users should access only what their profile permits; personal data fields (name, e-mail, phone) are flagged as personal data.
NFR-03	Usability	Forms should be simple, with clear labels, help text and pick lists that make requests quick to complete on desktop or mobile browsers.
NFR-04	Maintainability	Forms, picklist values and workflows should be easy to modify from the Zoho Creator builder.
NFR-05	Auditability	Each record should keep a unique autonumber ID and a reported date where applicable.
NFR-06	Availability	Access depends on platform availability and network connectivity.
NFR-07	Performance	Reports and dashboard charts should read directly from form data without custom queries.

Table 3.3: Non-Functional Requirements

3.4 Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator low-code application platform
Scripting	Deluge for validation, auto priority, feedback classification and e-mail actions
Client	Desktop, laptop or smartphone with a modern browser and a stable internet connection
Browser	Current version of Google Chrome, Microsoft Edge or Mozilla Firefox
Development	Zoho Creator form, report, page and workflow builders
E-mail	Zoho Creator mail service (sendmail) with a configured sender

Table 3.4: Hardware and Software Requirements

3.5 Use Case Model

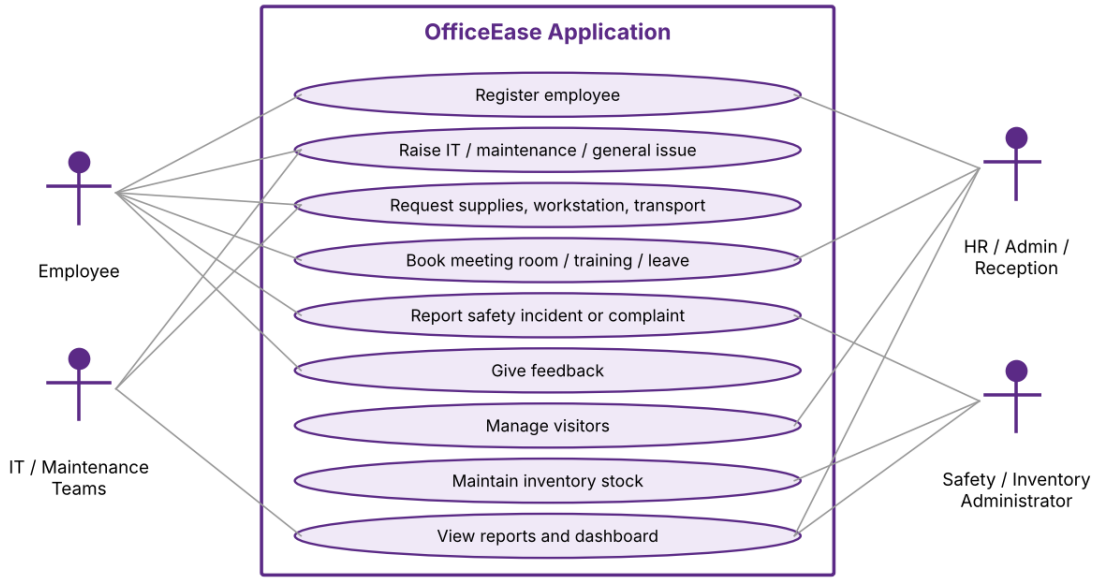


Figure 3.1: *Primary actors and the OfficeEase application*

Use Case	Precondition	Main flow	Result
Raise IT request	Employee knows the Employee ID.	Enter issue title, type, description, priority and attachment; submit.	Record stored; IT support team receives an e-mail alert.
Raise general issue	Employee has an issue to report.	Select category, describe the issue, enter location and submit.	Priority is assigned automatically from the category.
Request leave	Employee has chosen the dates.	Select leave type, dates and reason; submit.	Entry accepted, or rejected if the end date is before the start date.
Book meeting room	Room required.	Select facility, date, time, attendees and equipment; submit.	Booking recorded; facility team receives an e-mail.
Report safety incident	Incident has occurred.	Enter type, severity, location, injury details; submit.	Record stored; safety team receives an urgent e-mail.
Update stock	Item exists in inventory.	Enter available and minimum stock levels; submit.	Record saved; low stock alert e-mail is sent.

Table 3.5: Use Case Descriptions

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The architecture separates the user-facing layer, the automation layer and the stored application records. Employees and support teams interact with Zoho Creator forms, reports and the dashboard. Deluge workflows run on form events (validate and success) to enforce rules, assign priorities and send e-mail. Records are stored in the fifteen forms, and access is governed by Zoho Creator profiles and roles.

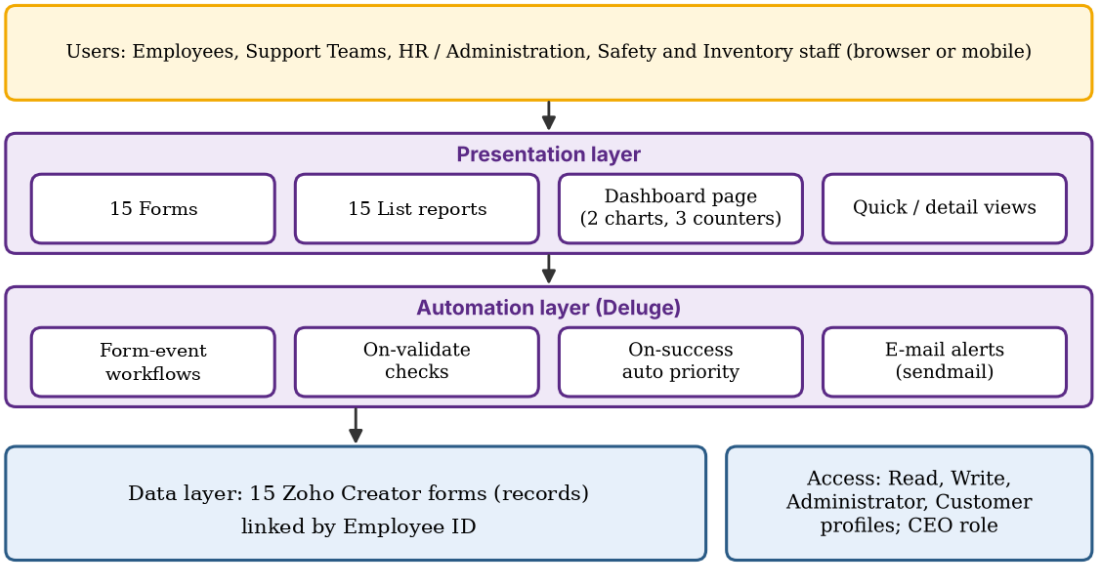


Figure 4.1: *Logical architecture of the OfficeEase application*

4.2 Data Model

Employee Registration acts as the master record. Every request form carries an Employee ID field, so that each request can be traced to the employee who raised it. In the current version this link is made by the Employee ID value entered in a text field; it is not a Zoho Creator lookup field. Replacing it with a lookup is listed under future enhancements.

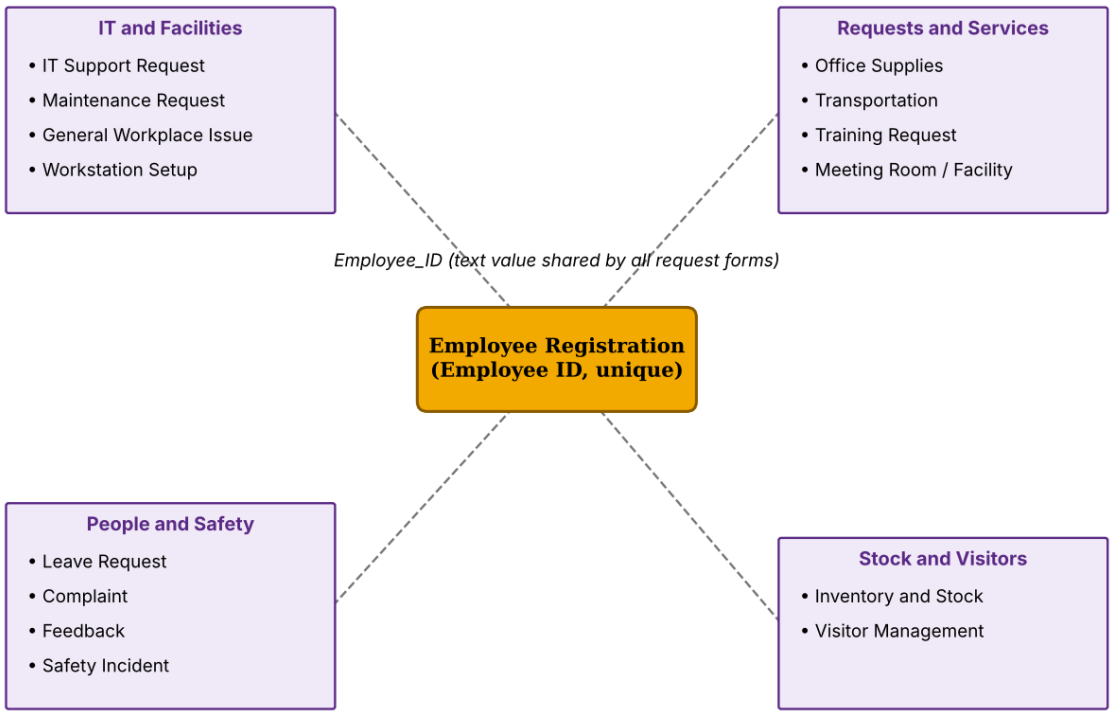


Figure 4.2: *Conceptual relationships between OfficeEase forms*

Group	Forms	Relationship / Purpose
Master data	Employee Registration	Unique Employee ID, e-mail and phone; referenced by value in every request form.
IT and facilities	IT Support, Maintenance, General Workplace Issue, Workstation Setup	Technical and physical workplace problems and equipment needs; most carry priority and attachments.
Requests and services	Office Supplies, Transportation, Training, Meeting Room and Facility	Employee service requests with dates, quantities and purposes.
People and safety	Leave, Complaint, Feedback, Safety Incident	Employee welfare, grievances, opinions and safety reporting.
Stock and visitors	Inventory and Supplies Stock, Visitor Management	Operational records maintained by the administrator and reception.

Table 4.1: Form Groups and Relationships

4.3 Form Design

All forms follow the same design pattern: an autonumber request ID (with its own start index), the Employee ID, a short title or category, a description, dates or quantities as needed, an optional attachment, and Submit / Update / Cancel actions. Mandatory fields are marked with “must have”, every field has help text that tells the user what to enter, and labels are placed to the left of the fields.

Field type	Used for	Examples
Autonumber	Unique request identifiers	IT Request ID, Complaint ID, Booking ID
Text / Textarea	Titles, descriptions and addresses	Issue Title, Problem Description, Pickup Address
Picklist / Radio buttons	Controlled values	Priority, Issue Category, Yes/No options
List / Checkboxes	Multiple selections	Equipment Required
Date / Time / Date-time	Scheduling and timestamps	Start Date, Pickup Time, Date Reported
Number / Currency (USD)	Quantities and amounts	Quantity, Number of Passengers, Amount
Email / Phone / Name	Contact details	Email Address, Phone Number, Full Name
Upload file	Photos and documents	Attachment, Supporting Document

Table 4.2: Form Design Groups

Field	Type	Mandatory	Values / Notes
IT_Request_ID	Autonumber	–	Starts at 10001
Employee_ID	Text	Yes	Registered employee ID
Issue_Title	Text	Yes	Short problem title
IT_Issue_Type	Picklist	Yes	Computer or Laptop, Software Problem, Printer Problem, Internet Connection, Network Problem, Email Problem, Login or Password, Keyboard or Mouse, Other
Problem_Description	Textarea	Yes	Problem and error message
Priority	Picklist	Yes	Low, Medium, High, Critical
Attachment	Upload file	No	Screenshot or photo
Date_Reported	Date-time	Yes	Date and time of report

Table 4.3: Field Specification – IT Support Request Form

Field	Type	Mandatory	Values / Notes
Issue_ID	Autonumber	–	Starts at 1
Employee_ID	Text	Yes	Registered employee ID
Issue_Title	Text	Yes	Short issue title
Issue_Category	Picklist	Yes	IT and Computer, Internet and Network, Electrical, Air Conditioning, Plumbing, Furniture, Cleaning and Housekeeping, Office Equipment, Safety and Security, Other
Issue_Description	Textarea	Yes	When it started and its effect
Specific_Location	Text	No	Floor, room or desk
Priority	Picklist	Yes	Low, Medium, High, Critical (set by workflow)
Attachment	Upload file	No	Photo or document
Date_Reported	Date-time	Yes	Date and time of report

Table 4.4: Field Specification – General Workplace Issue Form

4.4 Process and Workflow Design

Each request follows the same lifecycle. The employee opens a form and completes the mandatory fields. On submission the on-validate workflows run, and an invalid entry is rejected with an alert message. A valid entry is stored with an autonumber ID, after which on-success workflows assign a priority or send an e-mail to the responsible team. The team then reviews the record in the form's report, and the dashboard reflects the new counts.

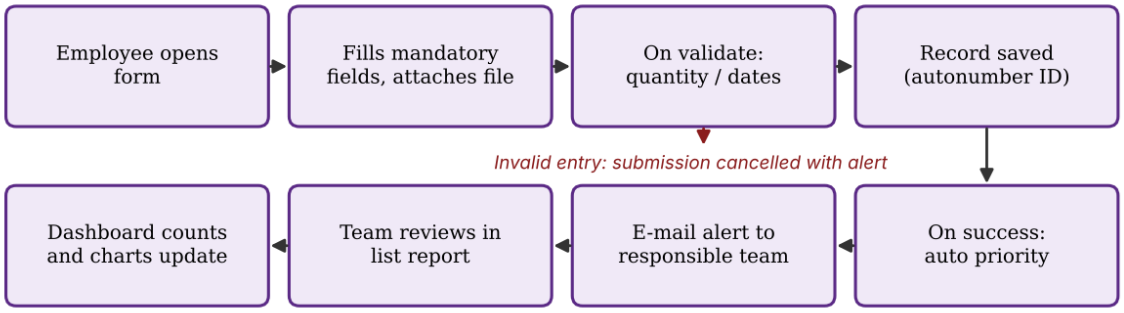


Figure 4.3: *Request lifecycle from submission to review*

4.5 User Interface and Navigation Design

The application opens in Zoho Creator with forms, reports and the Dashboard page listed in the navigation. Each report has a quick view (the key columns), a detail view (all columns) and record actions to view, edit, duplicate and delete. The Dashboard page uses a two-column layout with record counters at the top and charts below, and the pie chart has a Priority quick filter.

4.6 Access Control Design

Profile / Role	Type	Purpose
Administrator	Users profile	Full permissions on the application.
Write	Users profile	Write permission for all components.
Read	Users profile	Read-only permission for all components.
Customer	Customer portal profile	Default portal profile with add and view permission only; chat disabled.
CEO	Role	Role hierarchy entry whose users can access the data of other users.

Table 4.5: Access Control Design

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application OfficeEase was created in Zoho Creator with the date format dd-MMM-yyyy, a 24-hour time format and the UTC time zone. The exported application definition (version 1.0, generated on 01-Nov-2026) contains 15 forms, 15 reports, 1 page, 10 workflows and the sharing settings described in Chapter 4.

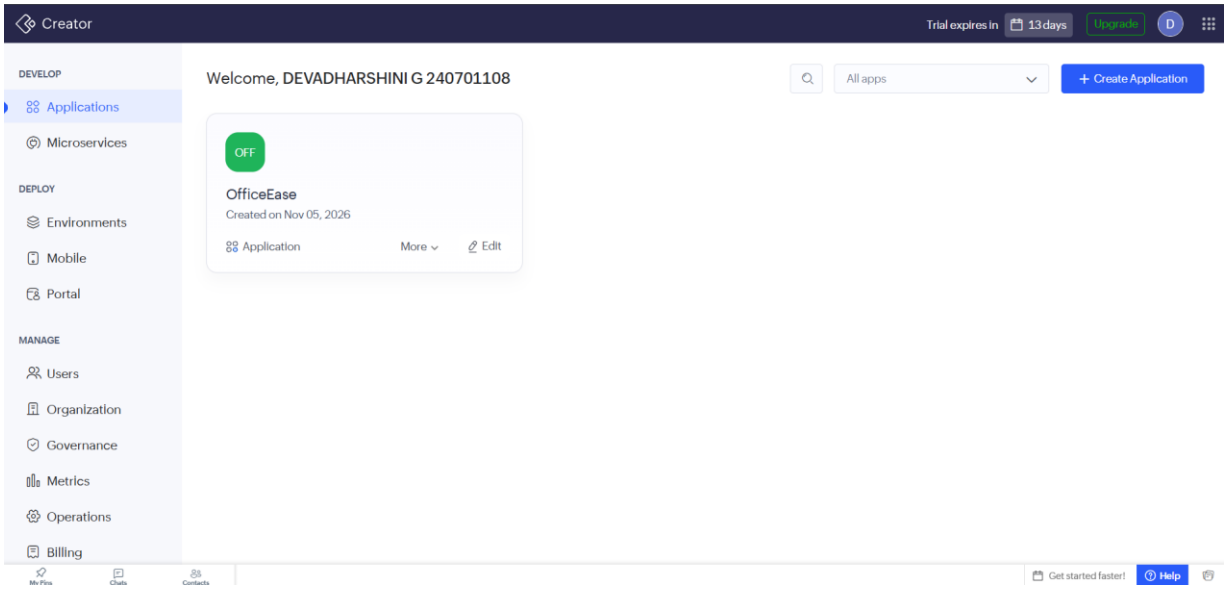


Figure 5.1: *OfficeEase Application in Zoho Creator*

5.2 Forms and Data Capture

Form	Link name	Fields	Autonumber start	Purpose
Employee Registration	Employee_Registration	10	–	Master record of employees: ID (unique), name, e-mail (unique), phone (unique), designation, department, location, joining date, manager, status.
IT Support Request Form	IT_Support_Request_Form	8	10001	IT problems with type, description, priority, attachment and date reported.
Maintenance Request Form	Maintenance_Request_Form	10	1000	Maintenance problems by category, equipment, location, priority and photo.
General Workplace Issue Form	General_Workplace_Issue_Form	9	1	General issues by category with location, attachment and automatic priority.
Office Supplies Request Form	Office_Supplies_Request_Form	10	101	Requests for stationery and consumables with quantity, dates and priority.

Form	Link name	Fields	Autonumber start	Purpose
Employee Workstation Setup Request Form	Employee_Workstation_Setup_Request_Form	10	1654	Desk, chair, monitor and other equipment requests.
Meeting Room and Facility Request Form	Meeting_Room_and_Facility_Request_Form	10	1234	Room bookings with date, times, attendees and equipment.
Employee Transportation Request Form	Employee_Transportation_Request_Form	10	102030	Transport requests with addresses, date, time, passengers and return trip.
Employee Training Request Form	Employee_Training_Request_Form	6	8526	Training topic, reason, mode and preferred date.
Employee Attendance and Leave Request Form	Employee_Attendance_and_Leave_Request_Form	7	3642	Leave type, dates, reason and supporting document.
Visitor Management Form	Visitor_Management_Form	11	15900	Visitor details, purpose, person to meet, ID type, exit time and status.
Workplace Safety Incident Form	Workplace_Safety_Incident_Form	10	4569	Incident type, description, location, severity, injury and attachment.
Employee Complaint Form	Employee_Complaint_Form	9	7895	Complaints by category with confidentiality option and preferred resolution.
Employee Feedback Form	Employee_Feedback_Form	5	9753	Service category, satisfaction rating (1–5) and comments.
Inventory and Supplies Stock Form	Inventory_and_Supplies_Stock_Form	10	78945	Item, category, code, available and minimum stock, supplier and location.

Table 5.1: Forms Implemented

The Employee Registration form enforces uniqueness on Employee ID, e-mail address and phone number, and marks the name, e-mail and phone fields as personal data.

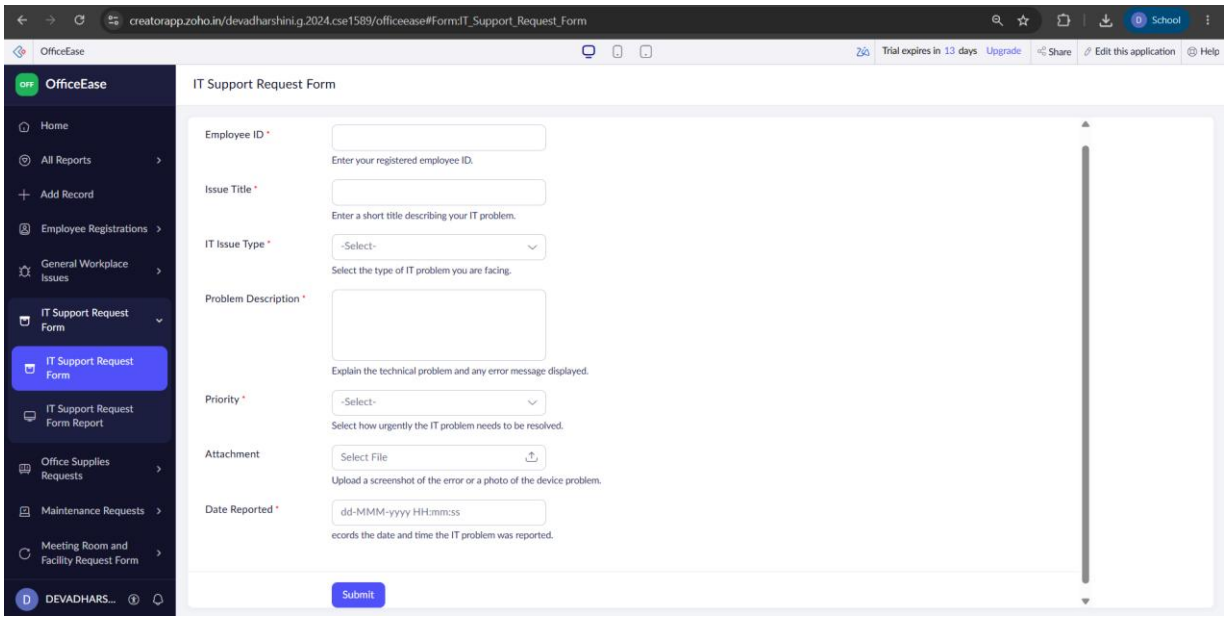


Figure 5.2: *IT Support Request Form in Live Mode*

5.3 Reports

Every form has a list report, and all fifteen are default reports showing all rows. Each report defines a quick view, a detail view and record actions. Appendix B lists the reports in detail.

Report	Source form	Type
All Employee Registrations	Employee Registration	List (default)
IT Support Request Form Report	IT Support Request Form	List (default)
All Maintenance Requests	Maintenance Request Form	List (default)
All General Workplace Issues	General Workplace Issue Form	List (default)
All Office Supplies Requests	Office Supplies Request Form	List (default)
All Employee Workstation Setup Requests	Workstation Setup Request Form	List (default)
Meeting Room and Facility Request Form Report	Meeting Room and Facility Request Form	List (default)
All Employee Transportation Requests	Transportation Request Form	List (default)
Employee Training Request Form Report	Training Request Form	List (default)
Employee Attendance and Leave Request Form Report	Attendance and Leave Request Form	List (default)
All Visitor Management	Visitor Management Form	List (default)
All Safety Incidents	Workplace Safety Incident Form	List (default)
All Complaints	Employee Complaint Form	List (default)
All Employee Feedback	Employee Feedback Form	List (default)
Inventory and Supplies Stock Form Report	Inventory and Supplies Stock Form	List (default)

Table 5.2: Reports Implemented

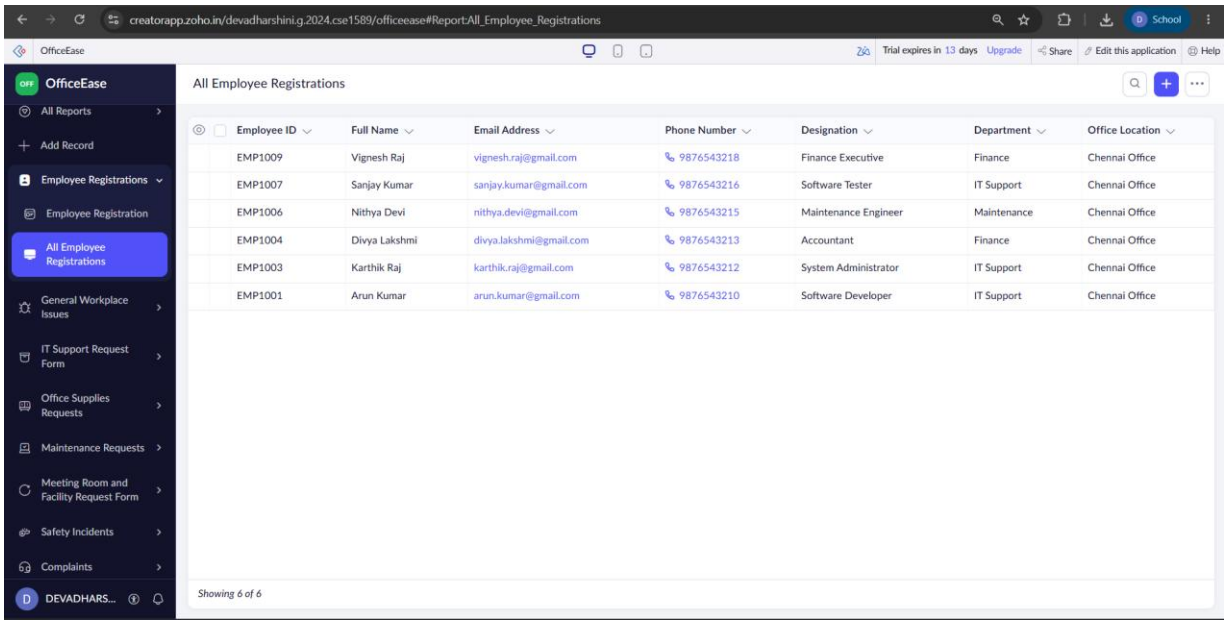


Figure 5.3: *Employee Registration Report*

5.4 Pages and Dashboards

Page	Components	Source data
Dashboard	Counter: Transportation Request	Employee Transportation Request Form (record count)
	Counter: Employee Attendance and Leave Request Form	Attendance and Leave Request Form (record count)
	Counter: “Complaints”	Workplace Safety Incident Form (record count)
	Column chart: Maintenance Category vs. count	Maintenance Request Form
	Pie chart: Issue Category vs. number of issues, with Priority quick filter and data labels	General Workplace Issue Form; linked to All General Workplace Issues report

Table 5.3: Pages Implemented

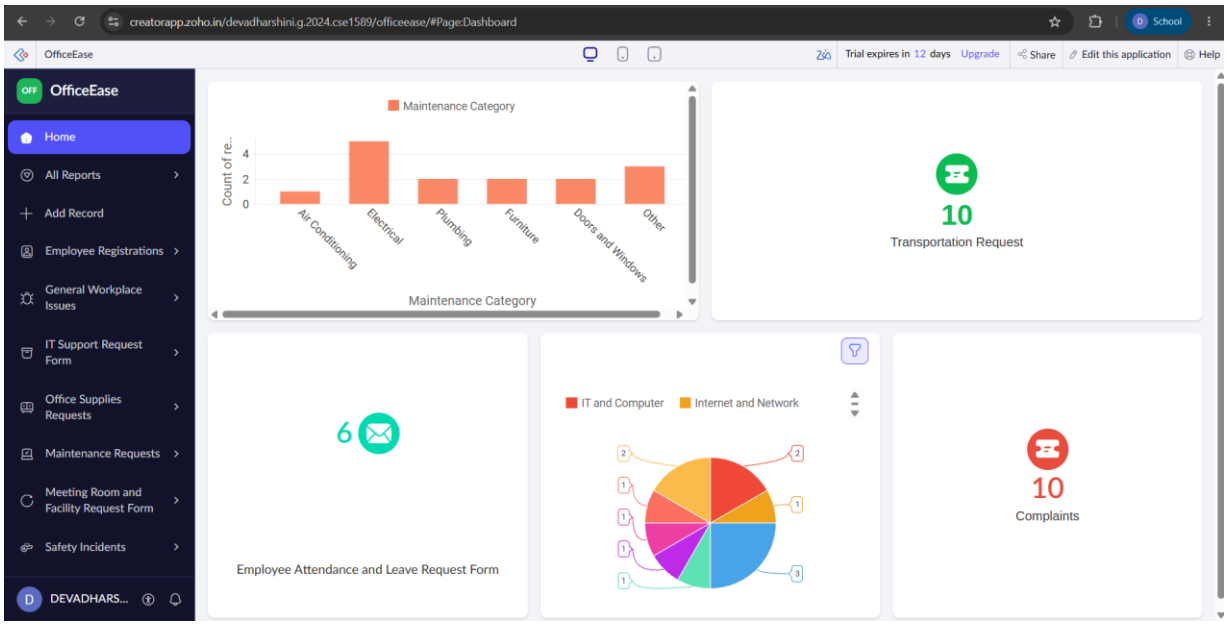


Figure 5.4: *Dashboard Page*

5.5 Deluge Scripting

Deluge scripts are used inside form workflows. They validate input, set the priority and classify feedback. Appendix C lists the logic. The priority rule maps the issue category to a priority, as shown in Figure 5.5.

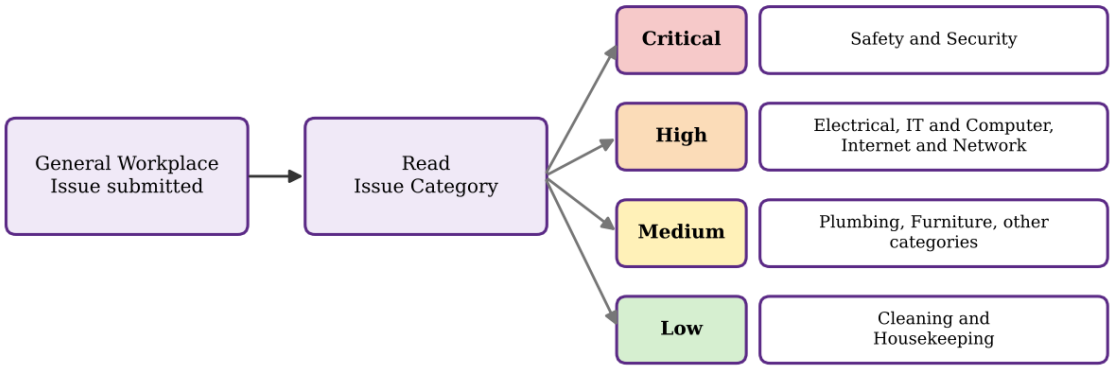


Figure 5.5: *Automatic issue priority assignment*

Script	Form	Event	Logic
Automatic Issue Priority Assignment	General Workplace Issue	On add – success	Sets Priority from Issue Category (Critical, High, Medium, Low).
Automatic Office Supplies Request Validation	Office Supplies Request	On add – validate	If Quantity ≤ 0: alert and cancel submit.
Automatic Leave Date Validation	Attendance and Leave Request	On add – validate	If End Date < Start Date: alert and cancel submit.

Script	Form	Event	Logic
Automatic Employee Feedback Classification	Employee Feedback	On add – success	Rating 4–5: “Positive Feedback”; 3: “Moderate Feedback”; otherwise “Negative Feedback” (info message).

Table 5.4: Deluge Logic Summary

5.6 Workflows and Automation

The application contains ten form workflows. Two run on validation, two run Deluge scripts on success and five send e-mail alerts using the sendmail task with the default e-mail template. One workflow (Workstation Equipment Quantity Validation) has been created but contains no actions in the exported application.

Workflow	Form	Trigger	Action
Automatic Issue Priority Assignment	General Workplace Issue	On add – success	Deluge script sets priority.
Meeting Room Booking Notification	Meeting Room and Facility Request	On add – success	E-mail to the facility team.
Workplace Safety Incident Alert	Workplace Safety Incident	On add – success	Urgent e-mail to the safety team.
Automatic Office Supplies Request Validation	Office Supplies Request	On add – validate	Rejects zero or negative quantity.
Automatic Leave Date Validation	Attendance and Leave Request	On add – validate	Rejects end date before start date.
Automatic Employee Feedback Classification	Employee Feedback	On add – success	Classifies feedback by rating.
Low Inventory Stock Alert	Inventory and Supplies Stock	On add or edit – success	E-mail to the inventory administrator.
Workstation Equipment Quantity Validation	Workstation Setup Request	On add – success	No actions configured.
Maintenance Request Alert	Maintenance Request	On add – success	E-mail to the maintenance team.
IT Support Request Notification	IT Support Request	On add – success	E-mail to the IT support team.

Table 5.5: Workflow Summary

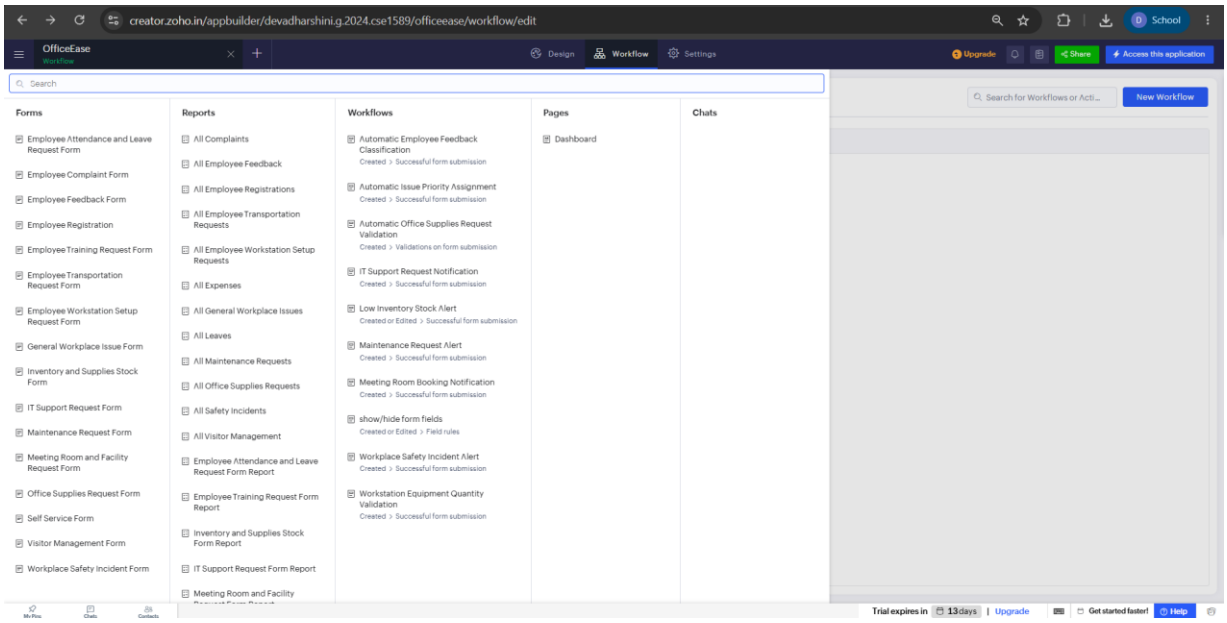


Figure 5.6: *Workflow list of the OfficeEase application*

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing follows the structure of the application. Forms are tested in live mode for mandatory fields, picklist values and attachments. Validation workflows are tested with valid and invalid inputs. On-success workflows are tested by submitting records and checking the stored values or the e-mail received. Reports and the dashboard are checked after sample records have been added. Access is checked by opening the application with different profiles.

6.2 Test Cases

The test cases below were designed from the configured logic of the application. The Status column is to be completed with the actual result recorded during live testing.

ID	Scenario	Input	Expected result	Status
TC-01	Mandatory field check	Submit IT Support Request without Employee ID	Form is not submitted; field is flagged	To record
TC-02	Unique Employee ID	Register two employees with the same ID	Second record is rejected	To record
TC-03	Unique e-mail	Register a duplicate e-mail address	Record is rejected	To record
TC-04	Leave date validation	Start 10-Nov-2026, End 08-Nov-2026	Alert “End date cannot be earlier than start date.”; submission cancelled	To record
TC-05	Valid leave request	Start 10-Nov-2026, End 12-Nov-2026	Record saved with autonumber ID	To record
TC-06	Supplies quantity validation	Quantity = 0	Alert “Quantity must be greater than zero.”; submission cancelled	To record
TC-07	Priority – safety	Issue Category = Safety and Security	Priority set to Critical	To record
TC-08	Priority – network	Issue Category = Internet and Network	Priority set to High	To record
TC-09	Priority – cleaning	Issue Category = Cleaning and Housekeeping	Priority set to Low	To record
TC-10	Feedback classification	Satisfaction Rating = 5 / 3 / 1	Positive / Moderate / Negative feedback message	To record
TC-11	IT support alert	Submit a valid IT Support Request	E-mail “New IT Support Request – OfficeEase” sent	To record
TC-12	Maintenance alert	Submit a valid Maintenance Request	E-mail to maintenance team sent	To record
TC-13	Safety incident alert	Submit a Safety Incident	E-mail “Urgent Workplace Safety Incident – OfficeEase” sent	To record
TC-14	Low stock alert	Add or edit an inventory record	E-mail “Low Stock Alert – OfficeEase” sent	To record
TC-15	Dashboard	Add records to Maintenance and General Issue forms	Charts and counters reflect the new records	To record

Table 6.1: Test Cases

6.3 Debugging Deluge Scripts

Scripts were debugged by using alert messages for validation failures, info messages to display the classification result, and the workflow execution history in Zoho Creator to check whether each workflow ran and whether the e-mail action succeeded. Statements were kept short and one rule was tested at a time.

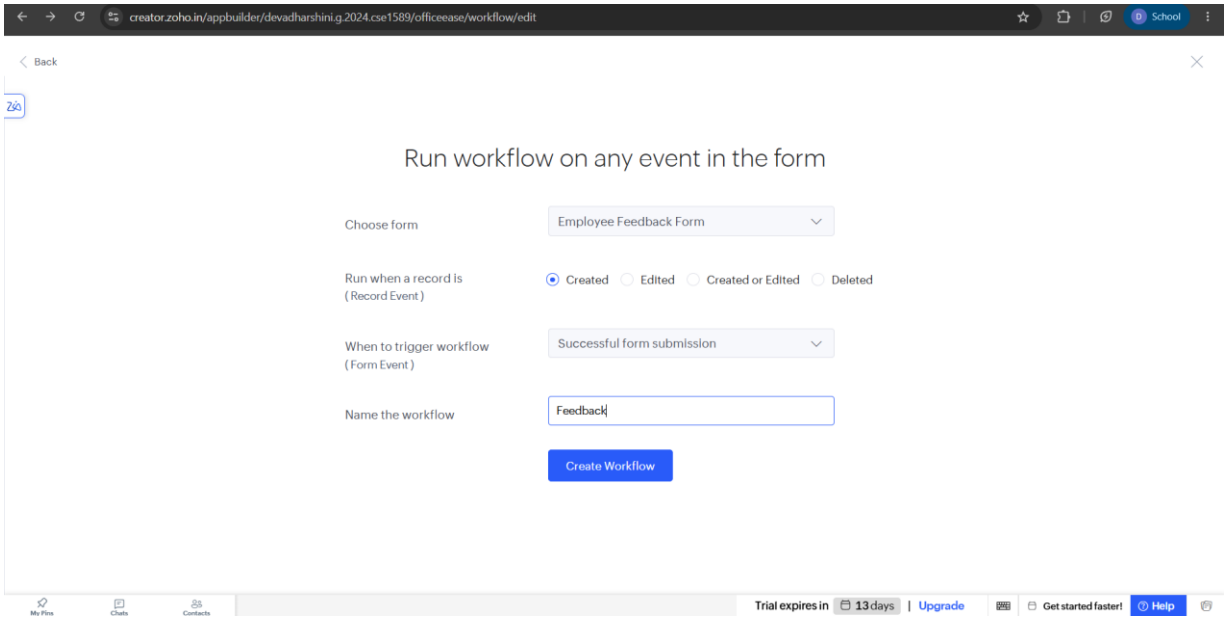


Figure 6.1: *Workflow execution history*

6.4 Observations and Validation Log

A review of the exported application produced the following observations. They are recorded honestly so that they can be corrected in the next version.

No.	Observation	Impact	Recommended action
1	The third dashboard counter is labelled “Complaints” but counts Workplace Safety Incident records.	Misleading label.	Rename to “Safety Incidents” or point it to the Complaint form.
2	Low Inventory Stock Alert runs on every add or edit without checking Available Quantity against Minimum Stock Level.	E-mail is sent even when stock is sufficient.	Add a condition: send only if Available_Quantity ≤ Minimum_Stock_Level.
3	Workstation Equipment Quantity Validation contains no actions.	No validation is applied.	Add a quantity > 0 check as in the supplies form.
4	E-mail alerts are sent to a single fixed address.	Teams other than the developer do not receive alerts.	Use team addresses or fetch them from a master record.
5	Priority is a mandatory field on the General Workplace Issue form, yet the workflow overwrites it automatically after submission.	Employees must choose a priority that is then replaced, which can confuse users.	Make Priority non-mandatory or hidden on the entry form and let the workflow set it.
6	Employee ID is a text field, not a lookup.	Typing errors can break the link to the employee.	Convert to a lookup to Employee Registration.

No.	Observation	Impact	Recommended action
7	Minor text issues: submit button on the Complaint form is labelled “Employee”; Maintenance alert subject ends with “OfficeEaseAdded”; one IT help text begins “ecords”.	Cosmetic.	Correct the labels.

Table 6.2: Observations and Validation Log

6.5 User Acceptance Testing

User acceptance testing is carried out by asking a few employees and support staff to perform the typical tasks in the user manual (Appendix D): register, raise an IT request, request leave, book a meeting room and view the dashboard. Feedback on clarity of labels, picklist options and report columns is collected and used to refine the application. Recorded UAT comments.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application is developed in the Zoho Creator builder (edit mode) and tested in live mode with sample records. Once testing is complete, the application is published for employees. Changes are made in the builder and tested again before being used in production.

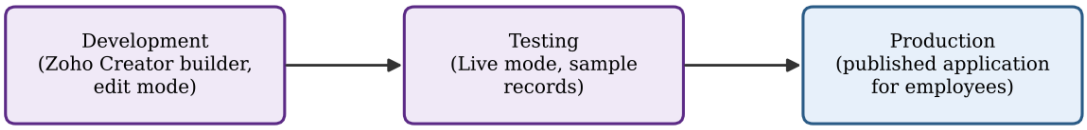


Figure 7.1: *Environments*

Item	Status
All forms created and field labels reviewed	Done
Workflows enabled and tested	Done in builder; execution evidence to be attached
E-mail sender and recipients verified	To be confirmed
Profiles and users assigned	To be confirmed
Application URL shared with users	[TO BE INSERTED]
Sample test data removed before go-live	To be done

Table 7.1: Deployment Checklist

7.2 Performance Optimisation

The application uses only built-in components. Reports show only the columns needed for the quick view; dashboard charts read directly from the form data; workflows are limited to a single form event each. Large attachments are optional so that submissions remain quick.

7.3 Security Features

Control	Implementation
Authentication and access	Zoho Creator sign-in; Read, Write, Administrator and Customer profiles
Role hierarchy	CEO role with access to other users' data
Data integrity	Mandatory fields; unique Employee ID, e-mail and phone number
Personal data marking	Name, e-mail and phone fields flagged as personal data
Input validation	On-validate scripts for dates and quantities
Transport security	Application is served over HTTPS by the platform

Control	Implementation
Confidentiality option	Complaint form lets the employee request confidential handling

Table 7.2: Security Controls Implemented

7.4 Governance and Administration

The administrator maintains users and profiles, reviews reports regularly and keeps picklist values current. Changes to workflows and forms should be made in the builder by authorised users and tested before use. Records containing personal data should be retained only as long as necessary.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The following scenario shows the application in use: an employee reports a network problem, the system assigns a priority and informs the team, and the team reviews the record.

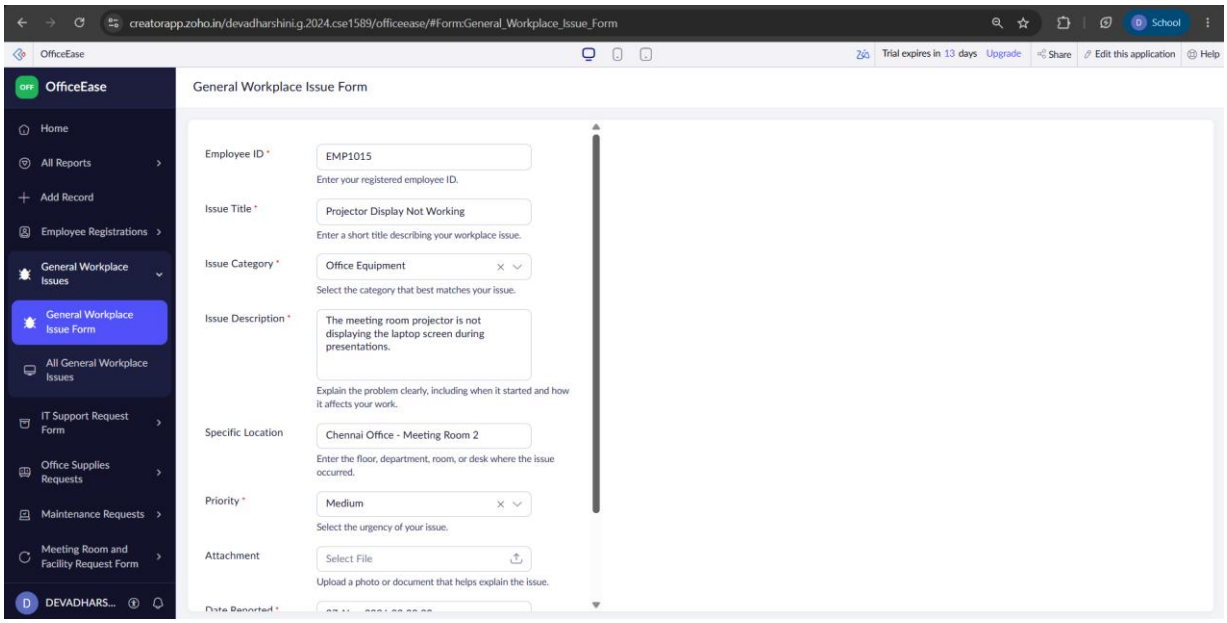


Figure 8.1: *Scenario Step 1 – Employee Raises an Issue*

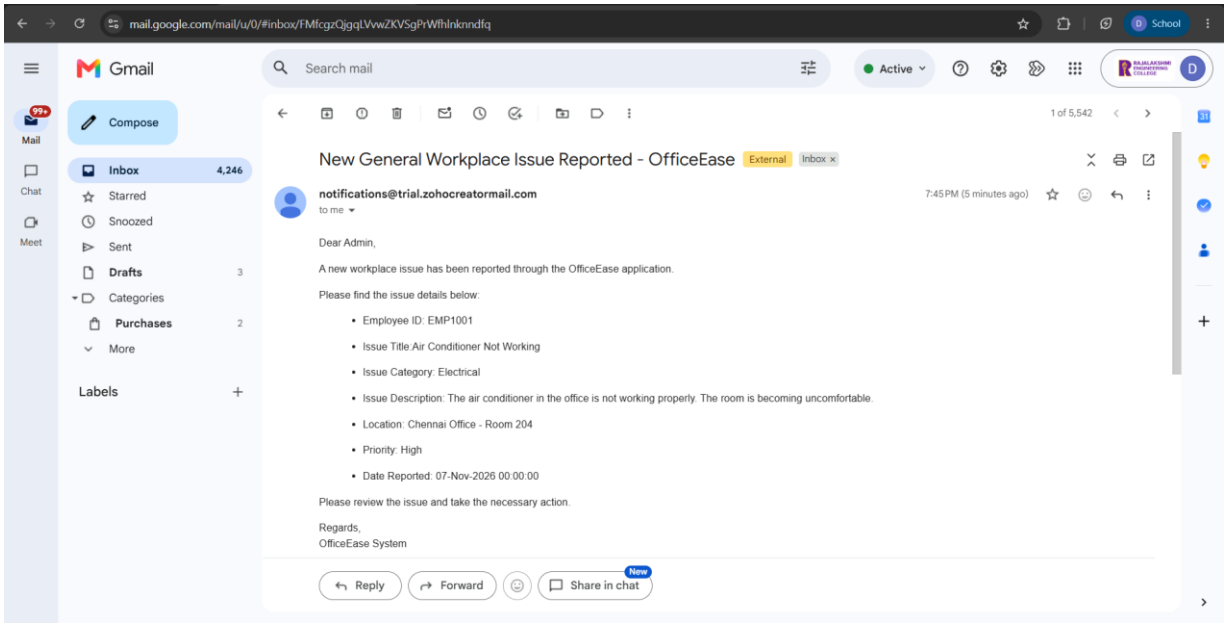


Figure 8.2: *Scenario Step 2 – Automatic Priority and E-mail Alert*

8.2 Achievement of Objectives

Objective	Achievement	Evidence
OBJ-01	Achieved	15 forms
OBJ-02	Achieved	Mandatory fields, picklists, help text, attachments
OBJ-03	Achieved	Leave date and supplies quantity validation
OBJ-04	Largely achieved	Auto priority, feedback classification, 5 e-mail alerts (some refinements pending, see Table 6.2)
OBJ-05	Achieved	15 reports and Dashboard
OBJ-06	Partly achieved	Profiles and role defined; record-level restrictions not configured

Table 8.1: Objective Achievement Mapping

8.3 Comparison with the Existing System

Aspect	Existing system	OfficeEase
Request capture	Phone, e-mail, verbal	Structured forms with mandatory fields
Validation	None	Automatic checks on dates and quantity
Priority	Decided informally	Automatic for general issues
Notification	Manual	E-mail alerts to responsible teams
Visibility	Manual consolidation	Reports and dashboard
Feedback	Not recorded	Rating with classification

Table 8.2: Comparison with the Existing System

8.4 Limitations of the Low-Code Approach

- Customisation is bounded by what the platform offers and by its plan limits.
- The application depends on the availability and plan limits of the platform.
- Scripts and permissions still need careful testing, as the observations in Chapter 6 show.
- The current version has no status tracking, approvals or scheduled jobs.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Forms, reports, charts and form rules were built without code. Validation, priority assignment and classification required short Deluge scripts, which is the low-code part of the work. A pro-code approach would be considered if the organisation needed complex integrations, a fully custom interface or heavy processing; for the present needs, low-code delivered a working application much faster.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

OfficeEase shows how a low-code platform can bring scattered workplace requests into one organised system. Fifteen forms capture employee requests, reports, complaints and feedback in a consistent way; ten workflows validate entries, assign priorities, classify feedback and alert the responsible teams; fifteen reports and a dashboard give visibility; and profiles and roles control access. The review also identified improvements, which are listed below.

9.2 Future Enhancements

- Replace Employee ID text fields with lookup fields to Employee Registration and display employee details automatically.
- Add Status, Assigned To and Resolution fields to every request, with a status lifecycle and closure date.
- Add approval workflows for leave, transportation and equipment requests.
- Add scheduled workflows to escalate overdue requests and send daily summaries.
- Make the low stock alert conditional and send e-mails to the correct team addresses.
- Complete the Workstation quantity validation and add a meeting room clash check.
- Restrict portal users to their own records and add more dashboard KPIs.
- Add SMS or push notifications and a mobile app.

9.3 Learning Outcomes and Career Relevance

Through this project I learned to design forms and data structures, use Deluge scripting for validation and automation, configure workflows and e-mail actions, build reports and dashboards, and apply profiles for access control. These skills are relevant to roles such as low-code developer, business application developer and automation analyst, where quick delivery of process applications is valued.

REFERENCES

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[2] Zoho Deluge Scripting Language Guide. https://www.zoho.com/deluge/help/

[3] Zoho Creator Help – Workflows. https://www.zoho.com/creator/help/workflows/

[4] Zoho Creator Help – Reports and Pages. https://www.zoho.com/creator/help/

[5] Zoho Creator Help – Customer Portal. https://www.zoho.com/creator/help/

[6] ServiceNow IT Service Management. https://www.servicenow.com/products/itsm.html

[7] Freshservice. https://www.freshworks.com/freshservice/

[8] OWASP Application Security Verification Standard. https://owasp.org/www-project-application-security-verification-standard/

APPENDIX A – FEATURE COVERAGE CHECKLIST

Feature	Implemented	Where
Forms and field types (text, picklist, radio, checkbox, list, date, time, date-time, number, currency, email, phone, name, upload, autonumber)	Yes	15 forms
Mandatory and unique fields	Yes	Registration and request forms
Help text and descriptions	Yes	All fields
Form rules (show / hide)	No	Not configured
On-validate scripts	Yes	Supplies, Leave
On-success scripts	Yes	Priority, Feedback
Send mail workflows	Yes	Five workflows
List reports with quick view and detail view	Yes	15 reports
Filtered report	No	Not configured
Dashboard page with charts and counters	Yes	Dashboard
Profiles (Read, Write, Administrator, Customer)	Yes	Sharing settings
Roles	Yes	CEO
Personal data marking	Yes	Registration form
Lookup fields and subforms	No	Employee ID is a text field
Scheduled workflows	No	Not configured
Approvals / Blueprint	No	Not configured
Custom functions	No	Not configured
Map views and integrations	No	Not configured

Table A.1: Feature Coverage Checklist

APPENDIX B – REPORTS AND PAGES CATALOGUE

B.1 Reports

Report	Main columns
All Employee Registrations	Employee ID, Full Name, Email Address, Phone Number, Designation, Department, Office Location, Date of Joining, Reporting Manager, Employment Status
All General Workplace Issues	Issue ID, Employee ID, Issue Title, Issue Category, Issue Description, Specific Location, Priority, Attachment, Date Reported
IT Support Request Form Report	IT Request ID, Employee ID, Issue Title, IT Issue Type, Problem Description, Priority, Attachment, Date Reported
All Office Supplies Requests	Supplies Request ID, Employee ID, Item Name, Other Item Name, Quantity, Purpose, Request Date, Required Date, Priority, Additional Notes
All Maintenance Requests	Maintenance Request ID, Employee ID, Issue Title, Maintenance Category, Problem Description, Affected Equipment, Specific Location, Priority, Attachment, Date Reported
Meeting Room and Facility Request Form Report	Booking ID, Employee ID, Meeting Title, Facility Type, Meeting Date, Start Time, End Time, Number of Attendees, Equipment Required, Additional Requirements
Other reports	All Safety Incidents, All Complaints, All Visitor Management, All Employee Transportation Requests, All Employee Workstation Setup Requests, Inventory and Supplies Stock Form Report, Employee Training Request Form Report, All Employee Feedback, Employee Attendance and Leave Request Form Report – each lists the fields of its form

B.2 Pages

Dashboard – one page with a two-column layout of counters and charts (see Table 5.3).

B.3 Dashboard Indicators

- Count of transportation requests.
- Count of attendance and leave requests.
- Count of safety incident records (labelled “Complaints”; see Table 6.2).
- Maintenance requests by category (column chart).
- General workplace issues by category with Priority filter (pie chart).

B.4 E-mail Alert Triggers

Alert	Form	Subject
IT support	IT Support Request	New IT Support Request – OfficeEase
Maintenance	Maintenance Request	Record Maintenance Request Received – OfficeEaseAdded
Meeting room	Meeting Room and Facility Request	New Meeting Room Booking Request – OfficeEase
Safety incident	Workplace Safety Incident	Urgent Workplace Safety Incident – OfficeEase
Low stock	Inventory and Supplies Stock	Low Stock Alert – OfficeEase

Table B.1: E-mail Alert Triggers

APPENDIX C – DELUGE LOGIC SUMMARY

C.1 Automatic Issue Priority Assignment

```
if(input.Issue_Category == "Safety and Security")
{ input.Priority = "Critical"; }
else if(input.Issue_Category == "Electrical")
{ input.Priority = "High"; }
else if(input.Issue_Category == "Internet and Network")
{ input.Priority = "High"; }
else if(input.Issue_Category == "IT and Computer")
{ input.Priority = "High"; }
else if(input.Issue_Category == "Plumbing")
{ input.Priority = "Medium"; }
else if(input.Issue_Category == "Furniture")
{ input.Priority = "Medium"; }
else if(input.Issue_Category == "Cleaning and Housekeeping")
{ input.Priority = "Low"; }
else
{ input.Priority = "Medium"; }
```

C.2 Office Supplies Quantity Validation (on validate)

```
if(input.Quantity <= 0)
{
    alert "Quantity must be greater than zero.";
    cancel submit;
}
```

C.3 Leave Date Validation (on validate)

```
if(input.End_Date < input.Start_Date)
{
    alert "End date cannot be earlier than start date.";
    cancel submit;
}
```

C.4 Feedback Classification (on success)

```
if(input.Satisfaction_Rating == "5" || input.Satisfaction_Rating == "4")
{ info "Positive Feedback"; }
else if(input.Satisfaction_Rating == "3")
{ info "Moderate Feedback"; }
else
{ info "Negative Feedback"; }
```

C.5 E-mail Alerts

The five alert workflows use the sendmail task with the sender zoho.adminuserid, the default e-mail template, a fixed recipient address, and a subject and message specific to the event (see Table B.1).

APPENDIX D – USER MANUAL

D.1 General Access

Open the authorized Zoho Creator application URL using the supported browser and sign in with your Zoho account. The menu lists the forms, reports, and Dashboard available to your profile.

D.2 Employee

- Open the required form (for example IT Support Request Form).
- Enter your registered Employee ID and complete all mandatory fields.
- Attach a photo or document if available and click Submit.
- A success message appears when the record is saved.

D.3 HR / Administration

- Add employees through Employee Registration with a unique ID, e-mail and phone number.
- Review leave, training, complaint and transportation reports.

D.4 IT, Maintenance, Safety and Inventory Teams

- Watch for e-mail alerts and open the matching report to see the details.
- Inventory staff add or edit records in the Inventory and Supplies Stock Form.

D.5 Administrator

- Assign users to profiles and roles.
- Review the Dashboard and reports regularly and keep picklist values up to date.

D.6 Troubleshooting

Problem	Likely cause	Action
Form will not submit	A mandatory field is empty	Complete the highlighted fields
“End date cannot be earlier than start date.”	End date before start date	Correct the dates
“Quantity must be greater than zero.”	Quantity is 0 or negative	Enter a positive quantity
No e-mail alert received	Alert is sent to a configured address	Check the recipient address and spam folder
Cannot see a report	Profile does not permit access	Ask the administrator

Table D.1: Troubleshooting

APPENDIX E – EVALUATION RUBRIC

Student Name	Devadharshini G	Register No.	2116240701108
Project Title	OfficeEase – Employee Workplace Issue and Request Management System	Date	

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.1: Evaluation Rubric for the Individual Project

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Table E.2: Performance Level Descriptors

Signature of Internal Examiner Name: _____	Signature of External Examiner Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

Student Name	Devadharshini G	Register No.	2116240701108
Department / Section	Computer Science and Engineering	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

Table F.1: Viva-Voce Questions and Responses

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Table F.2: Marks Summary

Examiners' Remarks:

_____

INTERNAL EXAMINER Name: _____	EXTERNAL EXAMINER Name: _____